

Oriel summer school

Principle agent problems

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Outline

1. Concept
2. Applications
3. Real world examples
4. Problem set

Introduction

- In this lecture we will explore how asymmetric information, moral hazard and adverse selection can all play a role in impeding collaboration between two parties. One who we will call the principal. The other agent

Introduction

- Principal-agent problems arise in various contexts.
- But the basic elements are
 1. Asymmetric information
 2. Misaligned incentives

Relevance of Principal-Agent Problems

- Principal-agent problems are relevant in various scenarios:
 - Corporate Governance
 - Politics
 - Healthcare
 - Financial Intermediation
- In these situations, conflicts of interest and information asymmetry can lead to inefficiencies and suboptimal outcomes.

Corporate Governance

- Shareholders delegate decision-making authority to managers.
- Managers might prioritize personal gains over maximizing shareholder value.
- Effective governance mechanisms are necessary to align interests and ensure accountability.

Politics

- Voters elect politicians to represent their interests.
- Politicians may not fulfil promises or act in the best interest of the constituents.
- Transparency, checks and balances, and electoral mechanisms are essential for effective governance.

Healthcare

- Patients rely on doctors for medical decisions.
- Doctors may be influenced by financial incentives or conflicts of interest.
- Regulatory frameworks and patient empowerment can mitigate risks and ensure quality care.

Financial Intermediation

- Depositors entrust banks with their funds.
- Banks may engage in risky investments or fraudulent activities.
- Regulatory oversight, transparency, and risk management are crucial for stability and trust in the financial system.

Real World Examples: Enron Scandal

Here the principals are shareholders and agent managers

- **Background:** Enron was once one of the world's largest energy companies before filing for bankruptcy in 2001 due to accusations of accounting fraud. The bankruptcy was at that point the largest in U.S. history.
- **Principal-Agent Problem:** The executives (agents) did not act in the best interest of the shareholders (principals). Instead, they manipulated financial statements to inflate company profits and hide losses. Shareholders were unaware of the manipulation (asymmetric information).
- **Consequences:** The true profits and losses eventually emerged, the stock price plunged and soon Enron was bankrupt. Shareholders lost their money, employees lost their retirement savings (tied to company stocks). Some of the executives were sentenced to prison.

Real World Examples: Wells Fargo Scandal

Here the agents are the firm's employees and the principals are the management.

- **Background:** In 2016, Wells Fargo employees opened millions of new bank accounts and credit cards in customers' names without their consent.
- **Principal-Agent Problem:** The employees (agents) were incentivized by high sales goals set by the management (principals), which led them to act against the best interest of the customers and commit fraud.
- **Consequences:** Wells Fargo paid billions in fines and lawsuit settlements. Huge reputational damage.

Two-Date Investment Model

- At time 0, an entrepreneur has an investment opportunity that costs I .
- After raising financing, the project starts and the entrepreneur makes an effort e . Outsiders **do not** observe effort
- At time 1, the project yields a cash flow X with probability e and 0 with probability $1 - e$.
- Effort is costs the entrepreneur $c(e) = ke^2$.

First-Best Outcome

(a) First-best outcome:

- If the entrepreneur can self-finance the project, what effort level would she choose?

Solution: Part (a)

The entrepreneur wants to maximize the Net Present Value (NPV) of the project, given by:

$$\text{NPV}(e) = -I + [e \cdot X + (1 - e) \cdot 0] - ke^2 = -I + eX - ke^2 \quad (1)$$

Where I is the investment cost and X is the cash flow from the project. To find the optimal effort level, we take the derivative with respect to e :

$$X - 2ke = 0 \quad \Rightarrow \quad e = \frac{X}{2k} \quad (2)$$

Solution: Part (a) (contd)

The entrepreneur undertakes the project if and only if the NPV, given the optimal effort level, is positive:

$$-I + \frac{X^2}{2k} - \frac{X^2}{4k} > 0 \quad \Rightarrow \quad 4kl < X^2 \quad (3)$$

Competitive Financing Market

Now suppose that the entrepreneur has no money and that:

- The entrepreneur needs to raise the entire amount I from outside financiers.
- The market for outside capital is competitive.
- Before the financing is raised, the entrepreneur owns all the equity in the firm.

Question (b)

Suppose the project has to be financed entirely with debt. Debt has a face value of B at time 1.

1. First, fix B . For a given B , what is the optimal effort of the entrepreneur? What is the value of debt?
2. What is the level of debt required to finance the project? Is it worth taking on the project? If so, what is the value of the firm, and how does it compare with part (a)?

Solution: Part (b)

Assuming that the debt value of B is enough to cover the investment cost I , the expected payoff to the entrepreneur in $t = 1$ will be:

$$e(X - B) - ke^2 \quad (5)$$

Whereas the expected payoff to the bondholders in $t = 1$ will be:

$$eB \quad (6)$$

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Given these expected payoffs, the entrepreneur maximizes her payoff with respect to the effort level e , hence:

$$X - B - 2ke = 0 \quad \Rightarrow \quad e = \frac{X - B}{2k} \quad (7)$$

Note that the effort level is now lower, the expected payoff to bondholders will be:

$$\frac{X - B}{2k} B = \frac{XB - B^2}{2k} \quad (8)$$

Solution: Part (b)

The value to the entrepreneur will be:

$$\frac{X-B}{2k} (X-B) - \frac{k}{4k^2} (X-B)^2 = \frac{(X-B)^2}{4k} \geq 0 \quad (9)$$

Always positive? Why? No risk for the entrepreneur anymore! The bondholder faces all the risk $\Rightarrow$ *moral hazard*

Solution: Part (b)

The bondholder must ensure that her net present value of investing is positive, i.e.:

$$-I + \frac{X - B}{2k} B \geq 0 \quad (10)$$

After substituting the effort level choice, this becomes:

$$\frac{XB - B^2}{2k} \quad (11)$$

Is this a tougher condition than one the entrepreneur is self-funded? Do we have more **credit rationing**? Let us play around by plugging in some simple values.

Solution: Part (b)

The project is worth undertaking for the entrepreneur for any positive level of B . The value of the firm will be the sum of the value of the inside equity owners plus the value of the debt (which is 0 as bondholders exactly break-even), i.e.:

$$V_{\text{firm}} = \frac{(X - B)^2}{4k} \quad (14)$$

Is the value of the firm lower now compared to internal funding?

Question (c)

Next, suppose the project has to be financed entirely with equity. Equity may be interpreted as a proportion α of the firm that needs to be sold to outside investors, with the entrepreneur retaining $1 - \alpha$.

1. For a given α , what effort level does the entrepreneur choose? What is the value of outside equity?
2. For what values of α is it worth taking on the project?
3. What is the level of outside equity (i.e., α) required to finance the project? Is it worth taking on the project?

Solution: Part (c)

(c) The payoff to the entrepreneur will be:

$$e(1 - \alpha)X - ke^2 \quad (15)$$

For this to be maximal, the entrepreneur chooses an effort level of:

$$e = \frac{(1 - \alpha)X}{2k} \quad (16)$$

Given this effort level, the outside equity holders get:

$$\alpha eX = \frac{\alpha(1 - \alpha)X^2}{2k} \quad (17)$$

Solution: Part (c)

The entrepreneur would only undertake the project if their expected payoff is greater than zero, i.e.:

$$\frac{(1-\alpha)^2 X^2}{2k} - \frac{(1-\alpha)^2 X^2}{4k} \geq 0 \quad (18)$$

Which is always satisfying.

The outside investors would only undertake this project if the investment amount is not bigger than their expected payoff, i.e.:

$$I \leq \frac{\alpha(1-\alpha)X^2}{2k} \quad (19)$$

Solution: Part (c)

The value of the firm is the sum of the payoffs to both the inside and outside equity holders, i.e.:

$$(1-\alpha) \frac{X}{2k} (1-\alpha) X - \frac{(1-\alpha)^2 X^2}{4k} + \frac{\alpha(1-\alpha) X}{2k} X = \frac{(1-\alpha^2) X^2}{4k} \quad (21)$$

To summarise

Now we compare all three forms of financing and show how principal-agent issues can lead to sub-optimal allocation.

Funding scheme	Internal	Debt	Equity
Firm value	$\frac{X^2}{2k}$	$\frac{(X-B)^2}{4k}$	$\frac{(1-\alpha^2)X^2}{4k}$
Effort	$\frac{X}{2k}$	$\frac{X-B}{2k}$	$\frac{(1-\alpha)X}{2k}$
Value to outsider	0	$\frac{X-B}{2k} B$	$\frac{\alpha(1-\alpha)X^2}{2k}$

Let us play around with different values and draw some insights.

What drives the results?

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Let us play around with different values and draw some insights.

What drives the results? Moral hazard from the entrepreneur (agent) who does not internalize the effect of his effort on the financier (principal). The financier cannot observe effort and cannot contract upon it. Asymmetric information $\Rightarrow$ sub-optimal allocation